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EU AI Act

Serious Incidents:

Predicting Impact of Fines

December 2025

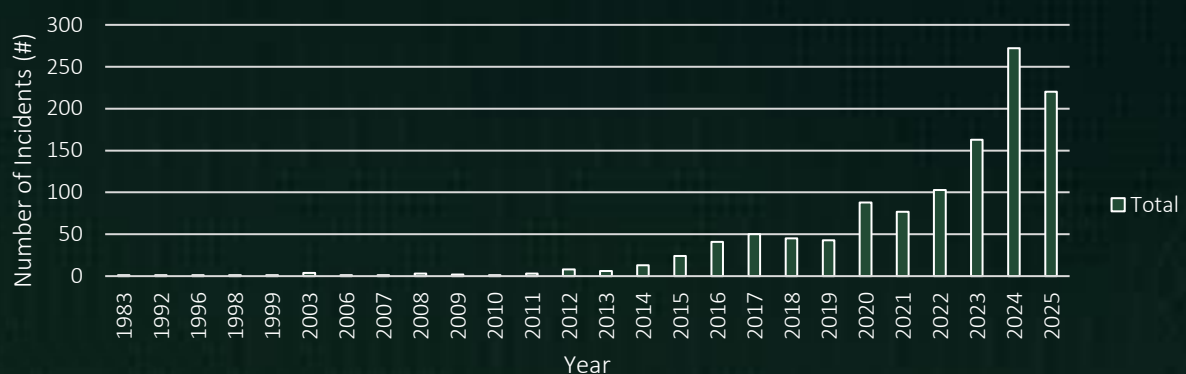
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Rapid acceleration of AI incidents highlights escalating operational, governance, and societal risks: organizations must prioritize monitoring, incident tracking, and risk-mitigation capacity

AI Incident Database: Weekly AI incident Snapshot (August 2025)



The chart shows a clear, exponential rise in AI-related incidents over time. For two decades—from the early 1980s through the mid-2000s—incident levels remained extremely low, with only isolated cases each year. Around 2015, however, the trend begins to shift sharply. By 2016–2018, annual incidents consistently reach the 40–50 range, in

dicating that AI systems had become sufficiently deployed in real-world contexts to generate recurring harms and failures. After 2020, the curve steepens dramatically. Incidents surge beyond 80 in 2020, exceed 100 by 2022, and then jump to more than 160 in 2023. The largest spike occurs in 2024, with incidents surpassing 270, followed by another elevated level of over 200 in 2025. This exponential pattern signals a structural, not incidental, change: AI systems are being deployed at scale across sectors, and the frequency of consequential failures is rising proportionally.

The trajectory suggests three key insights. First, AI incidents are no longer rare anomalies—they are systemic and increasing as model complexity and deployment density grow. Second, the steep post-2020 acceleration likely reflects the widespread adoption of generative AI, automated decision systems, and data-driven management tools, each introducing new failure modes and expanding the surface area for harm. Third, the volatility between 2024 and 2025 implies that as deployment expands, incident patterns may become more unstable and harder to predict, underscoring the need for continuous monitoring rather than one-time audits.

Constructively, the incident trend highlights the urgency of building organizational capacities for early detection, incident logging, root-cause analysis, and cross-functional response protocols. Regulators, firms, and civil-society actors all face a rapidly intensifying risk landscape—as shown by the chart's explosive growth curve—and must treat AI incident management not as a future concern but as an immediate operational priority.



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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

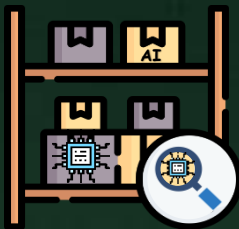
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

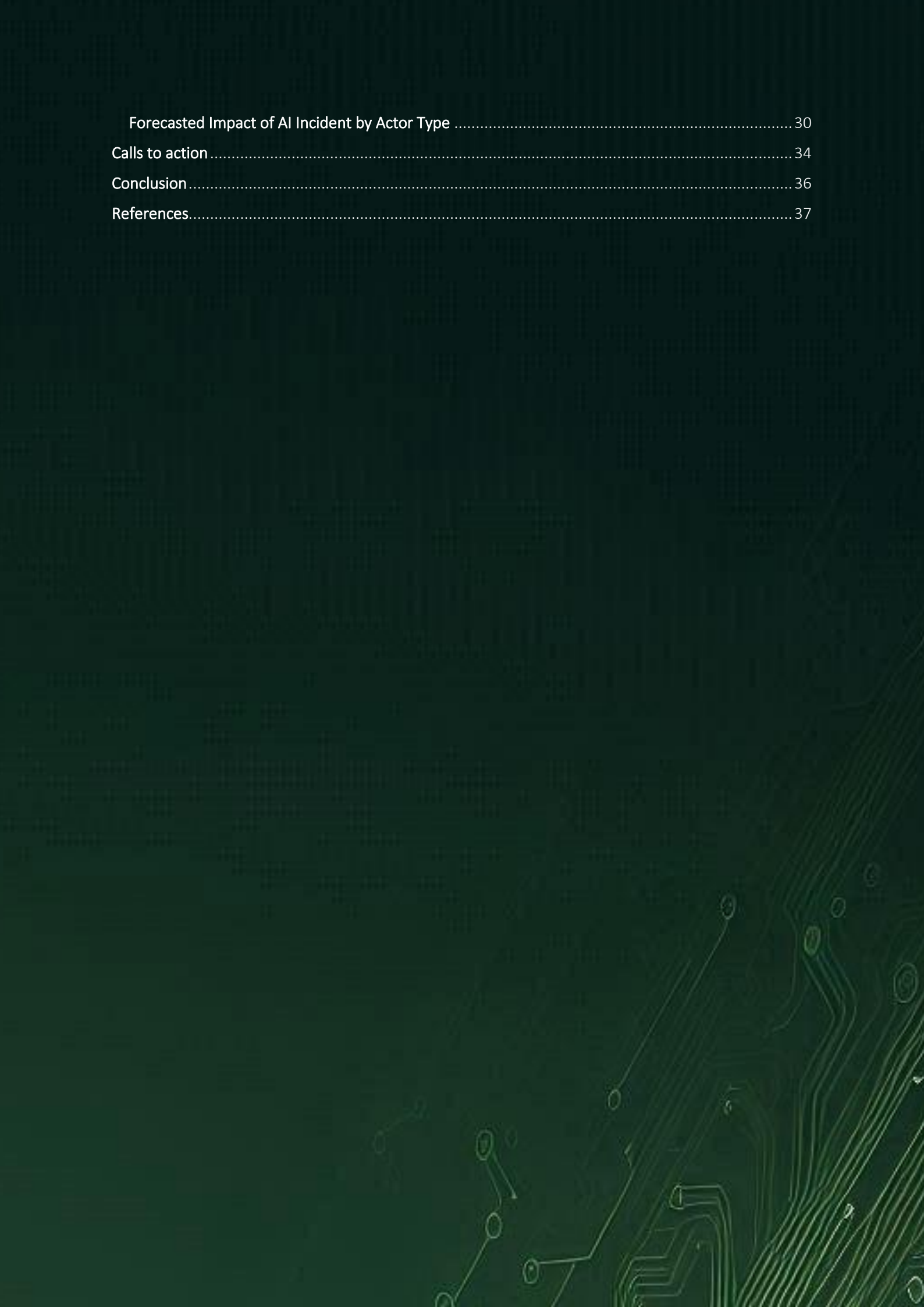
Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

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Introduction

As artificial intelligence continues to reshape industries, the regulatory battleground around serious AI incidents is rapidly intensifying — particularly where safety, public trust, and systemic risk intersect. New incident-reporting regimes under the EU AI Act demand that organisations treat AI failures not merely as technical faults but as events with legal, operational, and cross-border consequences; timely detection, documentation, and disclosure are now essential.

This report focuses squarely on AI incidents: the definition of a “serious incident,” mandatory reporting obligations and accelerated timelines for severe harms (including fatalities), and the practical expectations that market-surveillance authorities will place on providers and deployers. It explains what initial and follow-up reports should contain, who must notify authorities, and how regulators may respond once a report is submitted.

We map where harms are most likely to arise — from healthcare and finance to platforms, transport, and public administration — and show how governance gaps, opaque models, third-party dependencies, and weak change control turn localized failures into high-impact incidents that attract enforcement attention. The report also outlines operational controls that materially reduce incident likelihood and severity, such as live AI inventories, continuous model monitoring, evidence-preserving incident response workflows, and vendor governance.

Whether you are an AI developer, deployer, in-house counsel, or a regulator, this report delivers practical guidance for aligning operations with evolving incident-reporting expectations. AI & Partners prepares this analysis to help organisations anticipate enforcement priorities, harden controls, and demonstrate the governance maturity that regulators — and the public — will increasingly demand.

Best regards,

Sean Musch

Founder/CEO

AI & Partners



Key questions
being
asked about
insights on
forecasted AI
Incidents
fines drawn from
AI Incidents
Database



1. What constitutes a "serious incident" involving a high-risk AI system?

A serious incident involves any malfunction, performance issue, or behavior of a high-risk AI system that causes, or could potentially cause, harm to health, safety, or fundamental rights. This includes events such as incorrect decisions by the AI that impact individuals or society. The seriousness is determined by the potential or actual impact and aligns with criteria under Article 3, point (49)(b) of the AI regulation. Even if the harm is not immediate or obvious, if a causal link is suspected, it must be reported to ensure risks are promptly managed and mitigated. As the nature of AI inevitably brings a wide range of risks, specific risk management tailored to your organization is essential. Recognizing not only the known risks but also the unknown possibilities is vital. Our team can help identify, assess, and introduce these risks clearly—empowering your top managers to make better, more informed decisions.

2. When must providers report a serious AI incident?

Providers must report a serious incident as soon as they establish, or have reasonable grounds to suspect, a causal link between the incident and their high-risk AI system. This report must be made immediately, and in all cases, no later than 15 days after becoming aware of the incident. Timely reporting is essential for ensuring quick action by authorities and minimizing any ongoing or future risks associated with the AI system. Early notification also supports transparency and accountability in high-risk AI deployment.

3. Are there stricter reporting timelines for certain types of incidents?

Yes. For incidents involving widespread infringement or serious harm—such as large-scale violations of fundamental rights—providers must report them immediately and no later than two days after becoming aware. These are categorized under Article 3, point (49)(b). The accelerated timeline ensures that severe threats to public safety or rights are addressed quickly. Authorities rely on this rapid communication to issue alerts, impose temporary restrictions, or coordinate responses. The aim is to prevent harm from spreading and to uphold public trust in high-risk AI technologies.

4. What are the reporting requirements for AI incidents that result in death?

If an incident involving a high-risk AI system results in a fatality, the provider must submit a report immediately upon establishing or suspecting a causal link. Regardless of confirmation, the deadline is within 10 days of becoming aware of the incident. This provision prioritizes human life and requires urgent transparency. Authorities must be promptly informed so they can begin investigations, issue safety recommendations, or take regulatory action. Providers are also expected to support investigations and refrain from modifying the AI system unless coordinated with the relevant authorities.

5. Can providers submit a partial report initially?

Yes. To ensure prompt compliance, providers are allowed to submit an initial report with incomplete information. This preliminary report serves to alert authorities and initiate their response processes. A complete and detailed follow-up report must be submitted as soon as possible. This approach balances urgency with thoroughness, ensuring that regulators are informed quickly while giving providers the time needed to gather all necessary details. It also helps prevent delays that could lead to further harm or complications during the investigation phase.

6. What should the complete report include?

The complete report should detail the nature of the incident, the AI system's role, any suspected failures or malfunctions, a risk assessment, and any corrective actions taken or planned. It should also include data supporting the causal link, the timeline of events, affected parties, and any ongoing risks. This information enables authorities to determine the severity, coordinate responses, and decide whether additional oversight or product modifications are needed. The report should be factual, transparent, and in compliance with relevant regulatory templates or formats.

7. Are providers allowed to alter the AI system after an incident?

Providers must not alter the AI system involved in a serious incident without first informing the market surveillance authorities. Unauthorized changes may compromise investigations or obscure the causes of the incident. If a modification is urgently needed to prevent further harm, the provider must coordinate closely with authorities. Transparency and collaboration are essential for maintaining accountability and ensuring that any system updates are safe and legally compliant. The goal is to preserve system integrity during the assessment phase.

8. What responsibilities do providers have after reporting an incident?

After reporting, providers must investigate the incident thoroughly. This includes assessing the AI system's risks, identifying the root cause, and implementing corrective actions. They must cooperate fully with the authorities, supplying any requested data or support. Providers are expected to act in good faith to resolve the issue, prevent recurrence, and ensure the safety and reliability of their AI system. This responsibility extends until the system is deemed safe and compliant again by relevant authorities.

9. Who do providers report to when an incident occurs?

Reports must be made to the market surveillance authorities in the Member State(s) where the incident occurred. These authorities are responsible for reviewing the report, coordinating follow-up actions, and ensuring compliance with AI regulations. They may also share information with other national public authorities or bodies as specified in Article 77(1), especially if the incident poses broader public or cross-border risks. Centralizing communication ensures consistent and efficient regulatory oversight across the EU.

10. What happens after authorities receive an incident report?

Upon receiving a report, the market surveillance authority assesses the seriousness of the incident and may initiate an investigation. They may also inform other national authorities, especially if public health, safety, or fundamental rights are at stake. The authority may issue alerts, recommend the suspension of the AI system, or require further corrective actions. Providers are expected to fully cooperate throughout this process. This collaborative framework aims to quickly contain risks and promote responsible use of high-risk AI technologies.

Understanding insights on forecasted AI Incidents drawn from AI Incidents Database



Core Insights¹

Why will these harm types arise?

These harms arise because complex, poorly governed AI systems interact with sensitive data, legacy processes, and opaque models. Drift, data poisoning, vendor misconfigurations, inadequate testing, and weak change control amplify failures. Regulatory blind spots and rushed deployments in high-impact sectors convert technical faults into serious consequences.



Where will harms caused by AI Incidents likely arise?

Harms will surface where AI controls critical services: healthcare, finance, transport, public administration, law enforcement, and large platforms. Sectors with complex data flows, high automation, legacy systems, or heavy third-party dependencies face magnified risk. Cross-border platforms and supply-chain failures create systemic, cascading impacts across jurisdictions and long-term societal disruption globally.

What mitigating measures can firms deploy?

Deploy AI inventories, privacy-by-design, continual model monitoring, and rigorous pre-deployment impact assessments. Strengthen vendor due diligence, change-control, logging, and explainability; enforce human-in-the-loop oversight, rollback mechanisms, and incident response plans.

¹ Data from "Incidents" file drawn from Weekly Incident snapshot dated 2025-08-25 10:06 AM

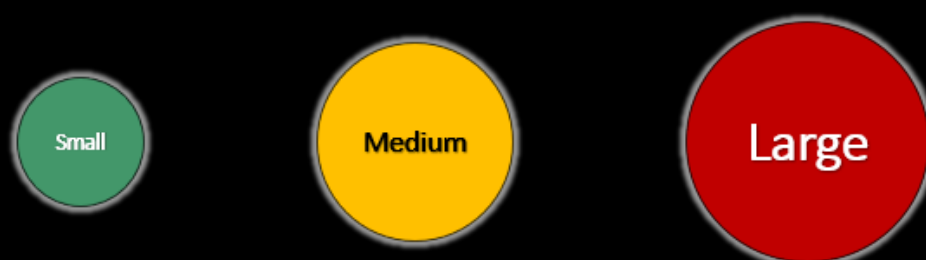
Assessing impact of forecasted AI Incidents drawn from AI Incidents Database



Forecasted Impact of AI Incidents by Sector



Legend



1. Accommodation & food service activities (hotels, restaurants, catering)

Impact type	Small	Medium	High
Operational	Minor booking/order glitches fixed by configuration or staff override.	Repeated service interruptions requiring centralized rollbacks and cross-site coordination.	Core operations disrupted across sites (kitchen or booking systems unusable).
Financial	Small refunds/discounts; negligible P&L effect.	Material remediation funds and lost weekend revenue at multiple sites.	Large revenue loss, bookings collapse, significant impairment to balance sheet.
Legal / Regulatory	Update guest terms & minor SLA clarifications.	Renegotiated vendor contracts; consumer agency investigations.	Fines, class actions, or license/permit threats from regulators.
Reputational	Short-term local complaints; limited social-media visibility.	Regional negative press, declines in review scores, temporary brand damage.	National scandal, prolonged negative media, brand trust erosion.
Safety / Human harm	Minor service-related annoyance; no injuries.	Local safety incidents (slip/fall) tied to automation, requiring remediation.	Serious injury from automated building controls or food-safety failures.
Data & Privacy	Small PII exposure to a handful of guests, remediated.	Significant guest-data leak requiring notifications and credit-protection offers.	Massive PII/PCI compromise with regulatory fines and forensic response.
Continuity / Supply chain	Single-supplier disorder corrected by manual procurement.	Repeated supply disruption forcing menu changes and supplier replacements.	Cascading supply disruptions across regions, extended closures.
Customer Experience & Trust	Isolated complaints; goodwill restorations suffice.	Widespread refunds/loyalty restorations required; measurable churn.	Mass cancellations, long-term loss of customer confidence.
Strategic / Market position	No strategic impact.	Competitive pressure to change vendor/technology choices.	Market-share loss; costly strategic repositioning required.

2. Administrative & support service activities (cleaning, facilities, security services)

Impact type	Small	Medium	High
Operational	Roster glitches or sensor false alarms fixed by rule updates.	Multi-site scheduling failures requiring manual intervention and overtime.	Systemic access-control failures causing suspended service delivery.
Financial	Minor overtime or remediation costs.	Contract penalties and compensation to major clients.	Loss of large contracts, major revenue decline.
Legal / Regulatory	Tighten change-management clauses in vendor agreements.	Client-driven indemnity demands and contract rework.	Debarment from public procurement; regulatory sanction.
Reputational	Local client complaints, contained.	Several clients publicize failures; market trust declines.	Industry-wide reputational damage and publicized lawsuits.
Safety / Human harm	Nuisance false alarms; no harm.	Safety incident causing non-life-threatening injury.	Serious injury or wrongful arrest due to automation misclassification.
Data & Privacy	Limited access-log exposure.	Sensitive access logs leaked across clients.	Deep privacy breach affecting many client sites and individuals.
Continuity / Supply chain	Single supplier swap needed.	Multiple vendor failures require procurement overhaul.	Supply chain collapse for critical contracts; inability to deliver.
Customer Experience & Trust	Complaint-driven retention work.	Contract renegotiations and loss of confidence among clients.	Long-term client churn and inability to win new bids.
Strategic / Market position	No change.	Reduced competitiveness for tendered contracts.	Forced exit from certain market segments.

3. Arts, entertainment & recreation (theatres, galleries, events)

Impact type	Small	Medium	High
Operational	Incorrect content tags or playlists; quick takedown.	Ticketing bot issues across events requiring refunds and reissue.	Major event cancellation due to AI-led safety failure.
Financial	Small refunds/compensation for attendees.	Large compensation for affected shows and lost sponsorships.	Catastrophic revenue loss, sponsor pullout, insolvency risk.
Legal / Regulatory	Update content-disclaimer and licensing clauses.	New licensing deals and indemnities, litigation from rights-holders.	Heavy fines for copyright/fraud, potential bans on certain practices.
Reputational	Niche criticism or artist complaints.	Widespread negative reviews and decline in attendance.	Cultural backlash, boycotts, institutional reputation damage.
Safety / Human harm	Minor crowding or discomfort incidents.	Injuries from mismanaged crowd flows tied to automation.	Mass injuries at large events caused by faulty AI crowd control.
Data & Privacy	Minor leakage of patron emails.	Ticketing database leaks exposing payment data.	Large-scale exposure of customer PII and payment details.
Continuity / Supply chain	One vendor failure affects a single show.	Multiple vendor failures disrupt season planning.	Long-term supply/partner collapse undermining event calendar.
Customer Experience & Trust	Single-show complaints; refundable.	Brand-level trust deterioration, membership cancellations.	Persistent avoidance by audiences and partners.
Strategic / Market position	Little to no effect.	Strained relationships with content partners.	Loss of major artist relationships and long-term competitive damage.

4. Defence (military, defence suppliers)

Impact type	Small	Medium	High
Operational	Test/simulation errors limited to training environments.	Degraded field performance requiring temporary stand-downs.	Unintended autonomous actions causing mission failures.
Financial	Minor R&D overruns.	Contractual penalties and accelerated remediation costs.	Program cancellation, massive budgetary write-offs.
Legal / Regulatory	Update internal RoE and procurement terms.	Export-control reviews and revised government contracts.	International legal scrutiny; treaty implications and sanctions.
Reputational	Limited internal confidence erosion.	Political questioning and media scrutiny.	Public outrage, parliamentary inquiries, long-term trust loss.
Safety / Human harm	No real-world harm; simulation only.	Non-fatal operational incidents affecting personnel.	Civilian casualties or friendly-fire events attributable to AI decisions.
Data & Privacy	Limited test-data leaks.	Sensitive program data leaked to adversaries.	Strategic compromise of classified systems and capabilities.
Continuity / Supply chain	Single-supplier vulnerability noticed.	Key supplier compromised, supply-chain delays.	Major weapon system supply collapse; capability gaps.
Customer Experience & Trust	N/A (government stakeholders)	Loss of confidence from allied partners.	Allies suspend cooperation and procurement.
Strategic / Market position	Minor R&D delays.	Loss of competitive edge in niche capabilities.	Strategic capability erosion with long-term defense ramifications.

5. Education (schools, universities, ed-tech)

Impact type	Small	Medium	High
Operational	Grading errors on isolated assignments.	Platform outages requiring manual grading and rescheduling.	Systemic assessment failures undermining degree validity.
Financial	Small remediation (make-up tests).	Refunds, remediation programs, vendor contract costs.	Loss of funding, enrollment drop, major fiscal impact.
Legal / Regulatory	Update academic policies and consent forms.	Data-protection investigations, contract rewrites.	Accreditation loss; national regulatory intervention.
Reputational	Local student complaints.	Public scandals around fairness/reliability of AI grading.	Long-term loss of institutional credibility and student trust.
Safety / Human harm	No physical harm; mental stress possible.	Student well-being impacted due to misclassification or exposure.	Harm to minors or vulnerable students through unsafe content exposure.
Data & Privacy	Single student record leak.	Sensitive student data shared with third parties.	Large-scale exposure of minors' data; regulator sanctions.
Continuity / Supply chain	Single ed-tech vendor downtime.	Multiple vendor outages disrupt teaching operations.	Nation-wide vendor failure affecting multiple institutions.
Customer Experience & Trust	Individual grievances handled.	Mass student and parent distrust; increased complaints.	Prospective students avoid institution; alumni trust declines.
Strategic / Market position	Minor reputational blip.	Need to shift ed-tech strategy and procurement.	Long-term decline in attractiveness; program closures.

6. Financial & insurance activities (banks, insurers, brokers)

Impact type	Small	Medium	High
Operational	Isolated model drift requiring rollback.	Widespread automation errors affecting client transactions.	Systemic trading or underwriting failure impacting market stability.
Financial	Small remediation or capital buffer use.	Large indemnities and client compensation; capital hit.	Systemic losses, runs, or insolvency risk for institution.
Legal / Regulatory	Update disclosures and internal model governance.	Major regulatory filings, new compliance obligations, heavy fines.	License revocation, criminal investigations, major regulatory restructuring.
Reputational	Local client complaints; manageable.	Significant brand damage; loss of high-value clients.	Systemic loss of public trust; industry-wide contagion effects.
Safety / Human harm	N/A typically; financial harm possible.	Customer financial distress from mis-advice.	Large-scale consumer harm leading to social/political consequences.
Data & Privacy	Small account-data mishap.	Leak of customer financial data, identity theft.	Massive PII leaks causing fraud and regulatory sanctions.
Continuity / Supply chain	Vendor model update delay.	Third-party failure causing operational degradation.	Chain reaction: clearing/custody or market infra failures.
Customer Experience & Trust	Individual remediation programs.	Widespread compensation and retention campaigns.	Customers withdraw funds; long-term loss of franchise value.
Strategic / Market position	Minimal.	Competitive repositioning to new risk controls.	Exit from markets, forced mergers, or break-ups by regulators.

7. Human health & social work activities (hospitals, clinics, care providers)

Impact type	Small	Medium	High
Operational	Minor triage/appointment scheduling errors corrected quickly.	Misdiagnosis patterns requiring widescale clinical review.	Widespread clinical error causing harm across facilities.
Financial	Small cost of readmissions or corrective care.	Large remediation, malpractice exposure, insurance cost increases.	Massive litigation, settlement costs and potential insolvency pressure.
Legal / Regulatory	Update clinical governance and consent templates.	Medical regulator investigations; contract rewrites with vendors.	Revocation of certifications, criminal investigations, sweeping reforms.
Reputational	Local patient complaints; contained.	Significant media coverage; patient trust declines.	Systemic loss of public confidence in healthcare providers.
Safety / Human harm	No harm; near-miss incidents.	Non-fatal adverse events tied to AI decisions.	Serious injuries/deaths attributable to AI-assisted care.
Data & Privacy	Single patient data exposure.	EHR leaks affecting many patients.	Large-scale health-data breach with regulatory fines.
Continuity / Supply chain	Single vendor issue (e.g., imaging provider).	Critical supplier failure causing treatment delays.	Collapse of critical clinical supply chain (medicines, devices).
Customer Experience & Trust	Individual patient remediation.	Patient transfers, complaints, drop in referrals.	Mass loss of faith in system; patients avoid certain providers.
Strategic / Market position	Limited.	Loss of competitive advantage in digital care.	Regulatory limits on AI use; program suspension.

8. Information & communication (telecoms, media, platforms, software)

Impact type	Small	Medium	High
Operational	Minor content-moderation misclassifications; patchable.	Platform outages or recommendation errors across geographies.	Major platform-wide outage, sustained service unavailability.
Financial	Small compensation or ad revenue loss.	Advertiser pullback and contract renegotiations.	Large revenue decline, advertiser exodus, market cap fall.
Legal / Regulatory	Tweak terms of service and moderation policies.	Regulatory investigations, new compliance mandates, cross-border legal complexity.	Forced structural change (e.g., mandated changes to algorithms), heavy fines.
Reputational	Localized user complaints.	National coverage around bias or misinformation.	Global loss of trust; user migration and brand collapse.
Safety / Human harm	Exposure to harmful content to small groups.	Dangerous content promoted to many users causing real harm.	Coordinated misinformation causing violence or mass harm.
Data & Privacy	Minor data-access bug.	Large dataset leak affecting millions.	Systemic privacy collapse with heavy fines and loss of user base.
Continuity / Supply chain	Vendor API outage affecting features.	Third-party service failure degrading core offerings.	Platform crippled by critical infrastructure failure.
Customer Experience & Trust	Temporary complaints; fixes accepted.	Decline in DAU/MAU, increased churn.	Massive user exodus, irreversible trust damage.
Strategic / Market position	Little effect.	Market share erosion in key demographics.	Forced divestiture, regulatory enforced breakup.

9. Law enforcement (police, corrections, public safety)

Impact type	Small	Medium	High
Operational	Misapplied facial-ID flags corrected with retraining.	Patrol-routing or triage errors causing missed responses.	AI-driven actions causing wrongful detention/arrest at scale.
Financial	Minor investigation costs.	Litigation costs and compensation; budget reallocation.	Massive settlements and funding crises for departments.
Legal / Regulatory	Internal policy clarifications and transparency commitments.	Judicial oversight changes and new use-of-force protocols.	Court rulings restricting or banning key AI uses; structural reforms.
Reputational	Limited community mistrust locally.	Public protests and scrutiny in multiple jurisdictions.	Systemic legitimacy loss; national protests and political fallout.
Safety / Human harm	Near-miss incidents; no injuries.	Non-fatal injuries due to misdirected resources.	Serious harm or death directly linked to AI-assisted policing.
Data & Privacy	Small data-handling error.	Leak of sensitive case data affecting investigations.	Mass privacy violations (e.g., illegal surveillance) triggering criminal probes.
Continuity / Supply chain	Vendor software bug affecting a department.	Multiple supplier failures affecting communications.	National-level operations degraded by critical vendor compromise.
Customer Experience & Trust	Decreased willingness to report crimes locally.	Significant community erosion of trust and cooperation.	Widespread refusal to cooperate with law enforcement.
Strategic / Market position	N/A (public body)	Loss of inter-agency trust and cooperation.	Restructuring of policing models and funding.

10. Manufacturing (factories, industrial production)

Impact type	Small	Medium	High
Operational	Production-line slowdown fixable by manual override.	Repeated machine miscalibrations reducing throughput.	Major quality-control failure producing unsafe products.
Financial	Minor rework costs.	Significant production losses, penalties for late delivery.	Massive recall costs, contract terminations, bankruptcy risk.
Legal / Regulatory	Update safety SOPs and product testing clauses.	Product-liability suits and regulatory inspections.	Mandatory product recalls, legal injunctions, heavy fines.
Reputational	Customer complaints and warranty claims.	Brand damage among key customers; lost contracts.	Global brand damage, long-term loss of OEM partnerships.
Safety / Human harm	No injuries; near-miss events.	Worker injuries due to automation error.	Fatal workplace accidents caused by autonomous systems.
Data & Privacy	Minor IP exposure.	Industrial espionage from leaked model/data.	Strategic loss of trade secrets and competitive IP.
Continuity / Supply chain	Single component delay.	Multiple supplier disruptions, inventory shortages.	Fragile supply chain collapse with global implications.
Customer Experience & Trust	Late shipments, small refunds.	Major customers switch suppliers.	Loss of key accounts and market exit.
Strategic / Market position	Short-term delays.	Need to redesign automation strategy.	Loss of competitiveness; forced divestment or acquisition.

11. Other (miscellaneous / uncategorised activities)

Impact type	Small	Medium	High
Operational	Localized process tweak.	Cross-function fixes and vendor changes.	Organizational paralysis for the activity in question.
Financial	Minor one-off expense.	Significant remediation costs for specific partners.	Potential insolvency for the activity's business unit.
Legal / Regulatory	Update internal policies.	Contract rework with affected parties.	Regulatory action if public harm occurs.
Reputational	Minimal.	Notable complaints in a niche market.	Severe reputational damage within niche/sector.
Safety / Human harm	No harm.	Local injuries or welfare issues.	Serious harm with broad consequences.
Data & Privacy	Small data issue.	Targeted data breach incidents.	Large-scale exposure affecting multiple stakeholders.
Continuity / Supply chain	Minor supplier swap.	Multiple supplier issues for niche inputs.	Supply collapse for the specific activity.
Customer Experience & Trust	Localized trust erosion.	Need for structured remediation.	Broad loss of trust among customers/partners.
Strategic / Market position	No effect.	Re-evaluation of strategic fit.	Exit or divestment of activity.

12. Professional, scientific & technical activities (legal, consulting, R&D, labs)

Impact type	Small	Medium	High
Operational	Minor report inaccuracies corrected by reviewers.	Rework of client-deliverables and loss of billable hours.	Systemic quality failures undermining core deliverables.
Financial	Small write-offs.	Client compensation and contract renegotiations.	Loss of major clients, revenue collapse for practice area.
Legal / Regulatory	Tighten disclaimers, engagement letters.	Overhaul of IP & liability clauses in client contracts.	Professional sanctions, litigation, loss of licenses.
Reputational	Client grumbles; reputational patching.	Publicized failures among major clients; thought-leader credibility loss.	Irreversible reputational damage; clients migrate permanently.
Safety / Human harm	Low or no physical risk.	Research errors causing harmful downstream effects.	Harmful scientific outcomes leading to public health/regulatory crises.
Data & Privacy	Minor dataset handling error.	Proprietary data leaks or IP misappropriation.	Loss of critical IP and client-trust due to data breaches.
Continuity / Supply chain	Single vendor delay.	Key vendor failure affecting project delivery.	Collapse of expert network or critical supplier access.
Customer Experience & Trust	Local client dissatisfaction.	Major client churn and reputational referrals decline.	Long-term erosion of client acquisition and retention.
Strategic / Market position	No immediate effect.	Need to re-evaluate service offerings.	Business model threatened; possible market exit.

13. Public administration (government departments, regulators)

Impact type	Small	Medium	High
Operational	Internal policy misapplication corrected internally.	Service delivery delays requiring inter-department fixes.	Critical public-service failure (benefits, licensing) affecting many citizens.
Financial	Minor remedial program costs.	Major budget reallocations to fix systems.	Large fiscal exposure, emergency funding, program collapse.
Legal / Regulatory	Update guidance and internal rules.	Legislative or regulatory amendments needed.	Judicial injunctions, legal reform, loss of public mandate.
Reputational	Local constituent upset.	Widespread political/media scrutiny.	Electoral fallout, crisis of confidence in institutions.
Safety / Human harm	Low risk.	Failures causing non-critical harm to citizens.	Large-scale harm due to automated public-service errors.
Data & Privacy	Small records-inconsistency.	Citizen-data exposure requiring mass notifications.	Massive privacy breach causing political crisis.
Continuity / Supply chain	Vendor hiccup in procurement.	Major supplier failure affecting critical services.	National-level failure of essential services.
Customer Experience & Trust	Local complaints.	Citizen lawsuits and reduced uptake of services.	Deep erosion of citizen trust in government digital services.
Strategic / Market position	Policy recalibration required.	Loss of program credibility.	Radical policy shifts and institutional restructuring.

14. Real estate activities (property management, brokerage, leasing)

Impact type	Small	Medium	High
Operational	Minor tenant-notification issues or access control hiccups.	Repeated building automation errors causing tenant disputes.	Major building-safety failures or mass evictions related to AI controls.
Financial	Small administrative costs.	Compensation to tenants and contract renegotiation.	Large claims, property valuation drops, investor lawsuits.
Legal / Regulatory	Update lease clauses and disclosure forms.	New contractual indemnities and compliance requirements.	Regulatory penalties, litigation, restrictions on automated property control.
Reputational	Local tenant dissatisfaction.	Market reputation hit among brokers and tenants.	Large investor and tenant flight from portfolios.
Safety / Human harm	No physical harm.	Local safety incidents caused by building automation.	Serious injury or fatality due to automated system failure.
Data & Privacy	Small access-log error.	Leak of tenant personal or financial data.	Broad exposure of tenant PII and lease data.
Continuity / Supply chain	Single vendor outage for building services.	Multiple vendor failures impacting building operations.	Widespread building operational collapse across portfolio.
Customer Experience & Trust	Tenant complaints handled case-by-case.	Increased vacancies and lease renegotiations.	Mass tenant departures and reputation damage.
Strategic / Market position	Minimal effect.	Need to revise proptech strategy.	Forced sale of assets or exit from markets.

15. Transportation & storage (airlines, rail, logistics, warehousing)

Impact type	Small	Medium	High
Operational	Route-optimization hiccups causing small delays.	Recurrent routing failures, missed deliveries, or flight delays.	Major safety incident (collision, derailment) linked to AI control.
Financial	Minor delay compensations.	Substantial penalties from late deliveries and lost contracts.	Multi-million recalls/cancellations, insurer actions, insolvency risk.
Legal / Regulatory	Update carrier terms and operational rules.	Regulator probes, revised safety certifications.	Grounding of fleets, revocation of operating certificates.
Reputational	Local customer complaints.	Shipper/customer trust decline; contract loss.	Public safety scare causing widespread avoidance of modes.
Safety / Human harm	No injuries; service disruption.	Injuries from automation-related incidents.	Fatal accidents attributable to AI routing/automation.
Data & Privacy	Minor routing-data leak.	Leak of shipment/manifest sensitive data.	Strategic compromise of logistics data causing national security concerns.
Continuity / Supply chain	Single-route disruption recoverable.	Network-scale disruptions causing regional paralysis.	Global supply-chain collapse amplified by automation failures.
Customer Experience & Trust	Delivery delays; refunds.	Large customer defection to competitors.	Long-term loss of contracts and market trust.
Strategic / Market position	Small operational tweak.	Rethink of automation strategy and vendor base.	Forced retrenchment or regulatory-imposed operating limits.

16. Unclassified (incidents not mapped to a sector)

Impact type	Small	Medium	High
Operational	Minor internal policy changes once classified.	Cross-functional remediation required.	Organizational-level response and restructuring.
Financial	Isolated small cost.	Material remediation burden for involved units.	Substantial liabilities with unknown downstream effects.
Legal / Regulatory	Internal policy updates.	External legal exposure once sector identified.	Major regulatory attention or litigation when severity high.
Reputational	Low if contained.	Public statements required and limited damage.	Broader public concern if incident reveals systemic failures.
Safety / Human harm	Isolated near-miss.	Confirmed harm affecting groups.	Severe harm provoking public inquiry.
Data & Privacy	Small dataset issue.	Significant data exposure requiring notices.	Large, multi-sector data compromise.
Continuity / Supply chain	Minor vendor issue.	Supplier ripple effects across units.	Extensive, multi-industry supply impacts.
Customer Experience & Trust	Limited to customers affected.	Wider dissatisfaction across product lines.	Loss of trust across the organization's customer base.
Strategic / Market position	No immediate effect.	Need strategic re-evaluation in affected area.	Potential existential threat to business model or unit.

17. Wholesale & retail trade (shops, supermarkets, wholesalers)

Impact type	Small	Medium	High
Operational	Pricing/checkout glitch at a single outlet.	Chain-wide pricing or stock-allocation errors.	System-wide inventory failures causing stockouts and hazards.
Financial	Minor discounting/refund costs.	Large lost sales and promotional failures.	Massive revenue shortfall, supplier/creditor actions.
Legal / Regulatory	Update return/refund policies and price-disclosure.	Regulatory complaints, consumer-rights suits.	Major consumer-fraud findings and fines.
Reputational	Local customer annoyance.	Chain-level negative reviews and media coverage.	Widespread boycott and deep brand damage.
Safety / Human harm	No injuries.	Customer injuries from automated checkout/hardware failures.	Major safety incidents at scale tied to automation.
Data & Privacy	Small POS data glitch.	Card-data or loyalty database leak.	Large-scale payment/PII breach with regulatory consequences.
Continuity / Supply chain	Single-supplier delivery issue.	Supplier network reconfiguration needed due to failures.	Retail supply collapse causing store closures.
Customer Experience & Trust	Local refunds; customers return.	Loyalty erosion and defections to competitors.	Permanent loss of customer trust and market share.
Strategic / Market position	Minimal.	Forced re-evaluation of automation/AI choices.	Long-term strategic repositioning or market exit.

Forecasted Impact of AI Incident by Actor Type

AI Developer

Bugfix/patch to model pipeline or CI; hotfix release.

Forced model rollback across customers; extended remediation sprints and support load.

Product-wide suspension, major release freeze, loss of operational capacity to maintain product.

Minor engineering hours and patch costs.

Compensation to customers, contractual penalties, higher support & insurance premiums.

Major indemnities, revenue clawbacks, loss of investor funding or insolvency for the provider.

Update developer T&Cs and internal compliance notes.

Requirement to insert stronger indemnities, provide audit logs and technical disclosures to

Regulatory enforcement against product (bans, mandated changes), litigation targeting

Negative developer forum/tech-blog mentions; limited PR handling.

Customer churn, loss of marquee customers; damage to hiring brand.

High-profile public scandal; loss of market trust and future sales pipeline.

Near-miss discovered in testing; internal process tightening.

Field incidents attributed to model error requiring public disclosure and litigation risk.

Harm or fatalities traced to algorithmic action — criminal/regulatory consequences and

AI Deployer

Reconfigure inference settings; manual overrides by ops.

Service-level outages across customers, multi-site rollback and operational overtime.

Production shutdowns, loss of key services, inability to meet customer SLAs.

Refunds/discounts for a subset of customers.

Large remediation funds, contract terminations with partners, increased insurance premiums.

Major revenue collapse, creditor action, bankruptcy risk for the deploying organization.

Amend deployment-specific clauses and disclosures.

Regulatory investigations, required compliance projects, contract renegotiation with

Fines, suspension of business licenses, forced cessation of the deployed service in jurisdiction(s).

Localized customer complaints and negative reviews.

Widespread negative press, partnerships paused.

Brand collapse, mass customer exodus, long-term market-share decline.

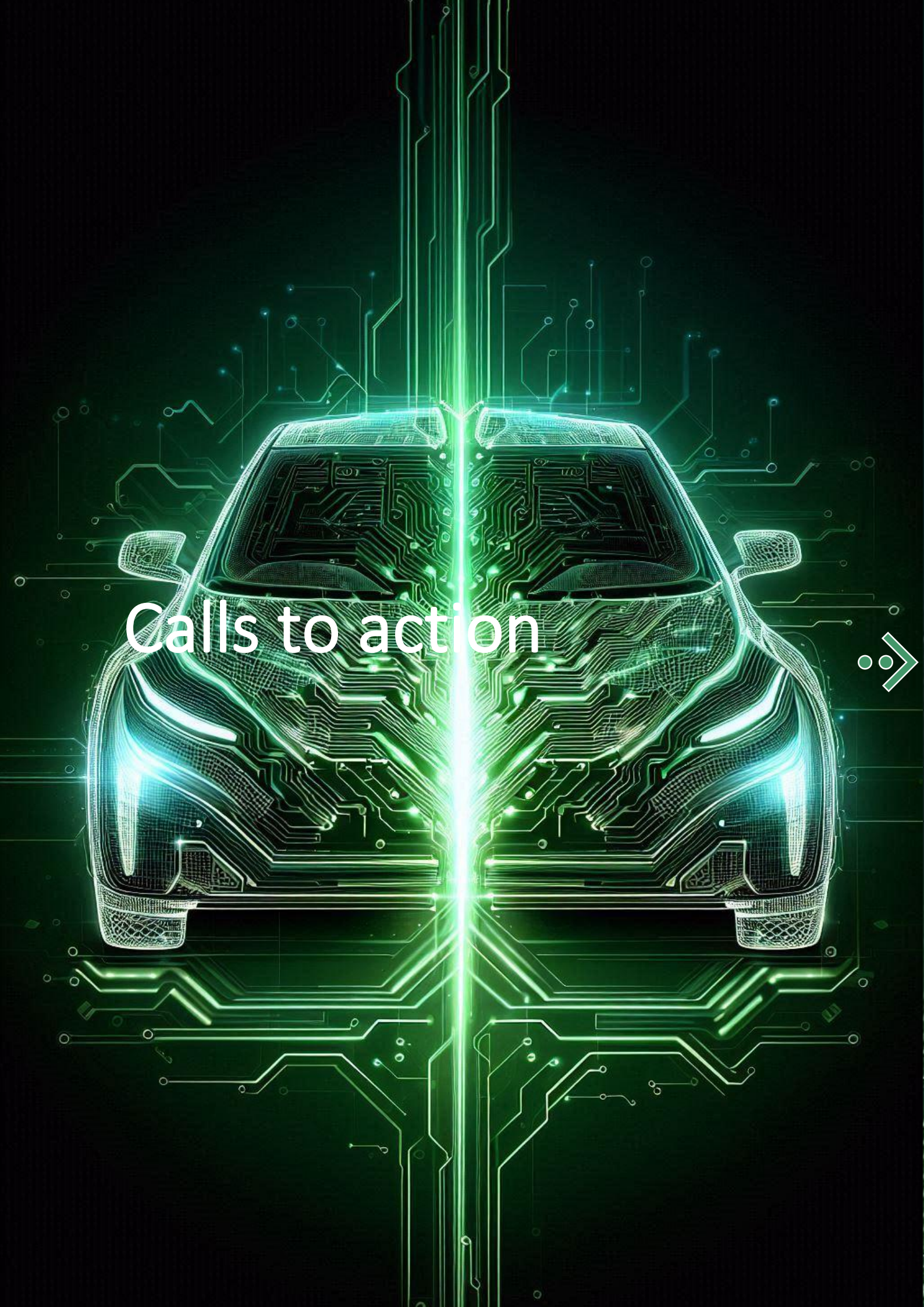
Minor user harm; immediate cessation of feature and localized remedy.

Multiple users harmed, compensation, and safety audits required.

Severe injuries/deaths leading to criminal/regulatory action, class action

Impact type	AI Developer (Small / Medium / High)	AI Deployer (Small / Medium / High)
Operational	Small: Bugfix/patch to model pipeline or CI; hotfix release.	Small: Reconfigure inference settings; manual overrides by ops.
	Medium: Forced model rollback across customers; extended remediation sprints and support load.	Medium: Service-level outages across customers, multi-site rollback and operational overtime.
	High: Product-wide suspension, major release freeze, loss of operational capacity to maintain product.	High: Production shutdowns, loss of key services, inability to meet customer SLAs.
Financial	Small: Minor engineering hours and patch costs.	Small: Refunds/discounts for a subset of customers.
	Medium: Compensation to customers, contractual penalties, higher support & insurance premiums.	Medium: Large remediation funds, contract terminations with partners, increased insurance premiums.
	High: Major indemnities, revenue clawbacks, loss of investor funding or insolvency for the provider.	High: Major revenue collapse, creditor action, bankruptcy risk for the deploying organization.
Legal / Regulatory	Small: Update developer T&Cs and internal compliance notes.	Small: Amend deployment-specific clauses and disclosures.
	Medium: Requirement to insert stronger indemnities, provide audit logs and technical disclosures to customers/regulators.	Medium: Regulatory investigations, required compliance projects, contract renegotiation with vendors.
	High: Regulatory enforcement against product (bans, mandated changes), litigation targeting the developer.	High: Fines, suspension of business licenses, forced cessation of the deployed service in jurisdiction(s).
Reputational	Small: Negative developer forum/tech-blog mentions; limited PR handling.	Small: Localized customer complaints and negative reviews.
	Medium: Customer churn, loss of marquee customers; damage to hiring brand.	Medium: Widespread negative press, partnerships paused.
	High: High-profile public scandal; loss of market trust and future sales pipeline.	High: Brand collapse, mass customer exodus, long-term market-share decline.
Safety / Human harm	Small: Near-miss discovered in testing; internal process tightening.	Small: Minor user harm; immediate cessation of feature and localized remedy.
	Medium: Field incidents attributed to model error requiring public disclosure and litigation risk.	Medium: Multiple users harmed, compensation, and safety audits required.
	High: Harm or fatalities traced to algorithmic action — criminal/regulatory consequences and severe liability.	High: Severe injuries/deaths leading to criminal/regulatory action, class action suits, loss of operating permission.

Impact type	AI Developer (Small / Medium / High)	AI Deployer (Small / Medium / High)
Data & Privacy	Small: Developer-side log exposure or minor PII leakage; rotate keys, patch.	Small: Small customer-data incident; notify affected users and patch.
	Medium: Customer data exposures requiring breach notification and contractual remediation.	Medium: Regulatory breach notifications, targeted lawsuits, mandated audits.
	High: Massive data breach tied to developer systems — heavy fines, loss of client base, legal suits.	High: Large-scale PII/PCI compromises, multi-jurisdictional fines, class actions, regulatory bans.
Continuity / Supply chain	Small: Third-party model/patch delay handled by contingency model.	Small: Substitute vendor or manual process triggered.
	Medium: Key dependency (library, API) broken causing degraded service for multiple deployers.	Medium: Multi-vendor failures forcing reprourement and service degradation.
	High: Critical supply-chain compromise (e.g., core model/weights IP lost), forcing major rebuild.	High: End-to-end supply-chain collapse (vendors, cloud, partners) causing sustained outages and inability to deliver services.
Customer Experience & Trust	Small: Customers notice minor quality regression; developer issues patch notes.	Small: Local customer annoyance resolved by compensation.
	Medium: Reputational requests for refunds, increased support burden and retention programs.	Medium: Measurable churn and negative NPS; marketing & CX remediation programs required.
	High: Loss of trust across customer base; long-term churn and broken contractual relationships.	High: Customers permanently abandon service; major loss of market credibility.
Strategic / Market position	Small: Slight slide in feature adoption; roadmap adjustments.	Small: Minor reprioritization of projects or vendor swaps.
	Medium: Loss of strategic partnerships, higher cost-of-sales, pivot in product strategy.	Medium: Need to re-evaluate digital strategy, switch suppliers, or revisit business model.
	High: Exit from market or loss of core product competitiveness; acquisition / insolvency pressure.	High: Forced strategic retreat from markets, divestment of business units, or overall business failure.



Calls to action



Build an AI Inventory Before Regulatory Deadlines

Automatically detect and register all internal and external AI systems, prompts, models, and vendors, creating a live inventory to ensure visibility, compliance with EU AI Act, and reduced enforcement risk across your organization quickly.



Continuously Monitor Models To Detect Performance Drift

Implement model-monitoring pipelines that identify drift, bias, or failures in production; trigger alerts, human-in-the-loop reviews, and rollback mechanisms to prevent harm, support incident reporting, and meet AI Act obligations promptly with clear audit trails.

Prepare Incident Response Plans And Reporting Templates

Create templates and workflows for rapid initial incident submission, follow-up investigations, evidence preservation, and regulator liaison; include timelines for severe incidents, fatality reporting, and coordinated vendor communication to comply with EU AI Act requirements.



Embed Privacy Design Across The AI Lifecycle

Adopt privacy-preserving techniques, data minimization, purpose limitation, and strong access controls from design through deployment; document decisions, DPIAs, and consent mechanisms to demonstrate lawful processing and reduce regulatory and reputational exposure across all systems.

Conclusion

The rise of targeted AI-incident regulation marks a pivotal shift in how safety, accountability, and system reliability are enforced across sectors. Incident-reporting regimes under the EU AI Act treat serious AI failures as not only technical faults but as regulatory events that require immediate notification, preservation of evidence, and cooperative engagement with market-surveillance authorities. These obligations create new operational imperatives for providers and deployers—especially around timely detection, causal analysis, and transparent reporting.

Regulatory impact will largely hinge on organisational readiness. Many firms still struggle to inventory deployed models, trace causal chains through complex supply chains, and keep robust change-control and logging practices that investigations demand. SMEs and smaller teams face particular challenges in resourcing these capabilities; without clear technical and governance investments they risk protracted investigations, mandated suspensions, and severe commercial consequences.

Encouragingly, early adopters across finance, healthcare, and technology are demonstrating that incident-aware governance is practicable and advantageous.

Organisations that embed continuous monitoring, human-in-the-loop safeguards, and evidence-preserving incident-response workflows reduce both the frequency and severity of incidents — and they shorten regulatory interactions when incidents do occur. These examples provide scalable patterns for others to follow.

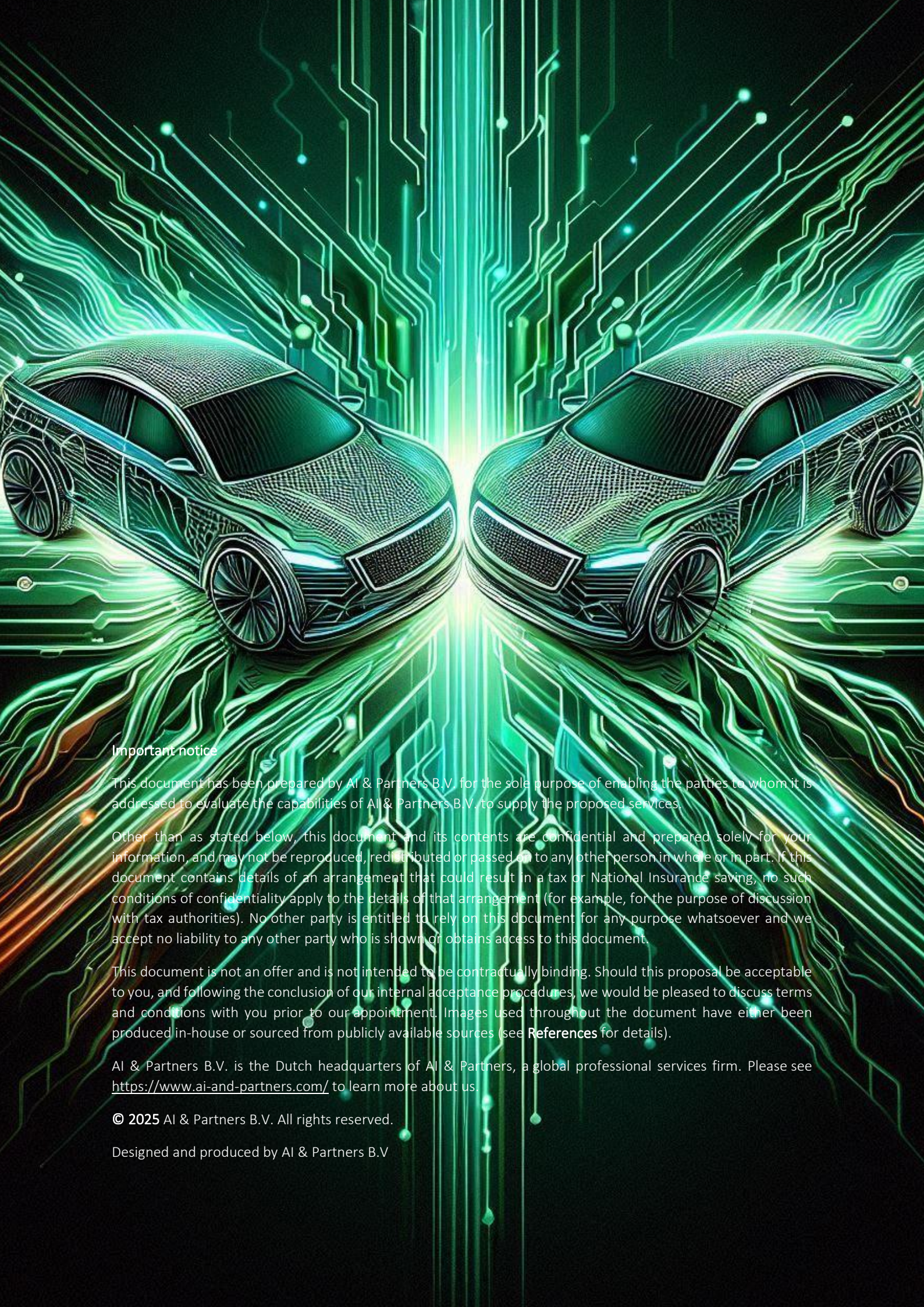
Practical preparedness pays dividends. Measures such as maintaining a live AI inventory, enforcing strict vendor due diligence, instituting model-monitoring pipelines, and codifying reporting templates materially lower the chance that an operational fault becomes a high-impact, reportable incident. They also enable faster remediation and clearer engagements with authorities, limiting disruption and reputational damage.

Ultimately, the EU AI Act's incident-reporting framework should be seen less as a compliance tax and more as a catalyst: it forces organisations to operationalise resilience and accountability into the lifecycle of AI systems. Firms that treat incident readiness as a strategic priority will be better positioned to protect users, sustain operations, and preserve trust — turning regulatory obligation into a source of competitive strength.

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